

```
mysql> -- Number Functions
mysql> SELECT ABS(-25) AS AbsoluteValue;
+-----+
| AbsoluteValue |
+-----+
|          25 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT POWER(salary, 2) AS SalarySquared FROM employees;
+-----+
| SalarySquared |
+-----+
| 9025142500.5625 |
| 7744088000.25 |
| 9801000000 |
| 5625037500.0625 |
| 11025000000 |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT MOD(emp_id, 2) AS EvenOdd FROM employees;
+-----+
| EvenOdd |
+-----+
| 1 |
| 0 |
| 1 |
| 0 |
| 1 |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT ROUND(bonus, 1) AS RoundedBonus FROM employees;
+-----+
| RoundedBonus |
+-----+
| 5000.3 |
| 4000.5 |
| 6000.8 |
| 3000.3 |
| 7000.0 |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT TRUNCATE(salary, 0) AS TruncatedSalary FROM employees;
+-----+
| TruncatedSalary |
+-----+
| 95000 |
| 88000 |
| 99000 |
| 75000 |
| 105000 |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT SQRT(81) AS SquareRoot;
+-----+
| SquareRoot |
+-----+
| 9 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> -- Aggregate Functions
mysql> SELECT COUNT(*) AS TotalEmployees FROM employees;
+-----+
| TotalEmployees |
+-----+
| 5 |
+-----+
1 row in set (0.04 sec)

mysql> SELECT SUM(salary) AS TotalSalary FROM employees;
+-----+
| TotalSalary |
+-----+
| 462001.50 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT AVG(salary) AS AverageSalary FROM employees;
+-----+
| AverageSalary |
+-----+
| 92400.300000 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT MIN(salary) AS MinSalary, MAX(salary) AS MaxSalary FROM employees;
+-----+
| MinSalary | MaxSalary |
+-----+
| 75000.25 | 105000.00 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT dept, AVG(salary) AS AvgDeptSalary FROM employees GROUP BY dept;
+-----+
| dept | AvgDeptSalary |
+-----+
| Management | 100000.375000 |
| R&D | 93500.250000 |
| Sales | 75000.250000 |
+-----+
3 rows in set (0.00 sec)
```

```
mysql> -- String/Character Functions
mysql> SELECT UPPER(emp_name) AS UpperName FROM employees;
+-----+
| UpperName |
+-----+
| ELON MUSK |
| LARRY PAGE |
| SUNDAR PICHAI |
| SATYA NADELLA |
| TIM COOK |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT LOWER(emp_name) AS LowerName FROM employees;
+-----+
| LowerName |
+-----+
| elon musk |
| larry page |
| sundar pichai |
| satya nadella |
| tim cook |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT LENGTH(emp_name) AS NameLength FROM employees;
+-----+
| NameLength |
+-----+
| 9 |
| 10 |
| 13 |
| 13 |
| 8 |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT CONCAT(emp_name, ' - ', dept) AS EmployeeDept FROM employees;
+-----+
| EmployeeDept |
+-----+
| Elon Musk - Management |
| Larry Page - R&D |
| Sundar Pichai - R&D |
| Satya Nadella - Sales |
| Tim Cook - Management |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT SUBSTR(emp_name, 1, 5) AS ShortName FROM employees;
+-----+
| ShortName |
+-----+
| Elon |
| Larry |
| Sunda |
| Satya |
| Tim C |
+-----+
5 rows in set (0.00 sec)
```

```
mysql> SELECT INSTR(emp_name, 'a') AS PositionOfA FROM employees;
+-----+
| PositionOfA |
+-----+
| 0 |
| 2 |
| 5 |
| 2 |
| 0 |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT REPLACE(city, 'o', '0') AS CityWithReplacedO FROM employees;
+-----+
| CityWithReplacedO |
+-----+
| New Y0rk |
| San Francisc0 |
| L0s Angeles |
| Chicag0 |
| B0st0n |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT LPAD(emp_name, 15, '*') AS LpadName FROM employees;
+-----+
| LpadName |
+-----+
| *****Elon Musk |
| *****Larry Page |
| *****Sundar Pichai |
| *****Satya Nadella |
| *****Tim Cook |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT RPAD(emp_name, 15, '*') AS RpadName FROM employees;
+-----+
| RpadName |
+-----+
| Elon Musk***** |
| Larry Page***** |
| Sundar Pichai** |
| Satya Nadella** |
| Tim Cook***** |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT TRIM(LEADING 'S' FROM 'SSMITHSS') AS LtrimExample;
+-----+
| LtrimExample |
+-----+
| MITHSS |
+-----+
1 row in set (0.03 sec)

mysql> SELECT TRIM(TRAILING 'S' FROM 'SSMITHSS') AS RtrimExample;
+-----+
| RtrimExample |
+-----+
| SSMITH |
+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT CURDATE() AS Today;
```

Today
2025-09-30

1 row in set (0.00 sec)

```
mysql> SELECT NOW() AS CurrentDateTime;
```

CurrentDateTime
2025-09-30 22:31:19

1 row in set (0.00 sec)

```
mysql> SELECT DATE_ADD(hire_date, INTERVAL 6 MONTH) AS SixMonthsAfterHire FROM employees;
```

SixMonthsAfterHire
2020-07-15
2019-09-22
2022-01-05
2019-05-30
2022-12-01

5 rows in set (0.00 sec)

```
mysql> SELECT DATE_SUB(hire_date, INTERVAL 1 YEAR) AS OneYearBeforeHire FROM employees;
```

OneYearBeforeHire
2019-01-15
2018-03-22
2020-07-05
2017-11-30
2021-06-01

5 rows in set (0.00 sec)

```
mysql> SELECT LAST_DAY(hire_date) AS LastDayOfHireMonth FROM employees;
```

LastDayOfHireMonth
2020-01-31
2019-03-31
2021-07-31
2018-11-30
2022-06-30

5 rows in set (0.00 sec)

```
mysql> SELECT CAST('1234.56' AS DECIMAL(10,2)) AS ToNumber;
```

ToNumber
1234.56

1 row in set (0.03 sec)

```
mysql> SELECT CAST(salary AS CHAR) AS SalaryAsText FROM employees;
```

SalaryAsText
95000.75
88000.50
99000.00
75000.25
105000.00

5 rows in set (0.00 sec)

```
mysql> SELECT DATE_FORMAT(hire_date, '%d-%M-%Y') AS FormattedHireDate FROM employees;
```

FormattedHireDate
15-January-2020
22-March-2019
05-July-2021
30-November-2018
01-June-2022

5 rows in set (0.01 sec)

```
mysql> SELECT STR_TO_DATE('15-02-2021', '%d-%m-%Y') AS ToDate;
```

ToDate
2021-02-15

1 row in set (0.00 sec)

```
mysql> SELECT TIMESTAMPDIFF(YEAR, hire_date, CURDATE()) AS YearsWorked FROM employees;
```

YearsWorked
5
6
4
6
3

5 rows in set (0.00 sec)

```
mysql> SELECT GREATEST(hire_date, '2020-01-01') AS GreatestDate FROM employees;
```

GreatestDate
2020-01-15
2020-01-01
2021-07-05
2020-01-01
2022-06-01

5 rows in set (0.03 sec)

```
mysql> SELECT LEAST(hire_date, '2020-01-01') AS LeastDate FROM employees;
```

LeastDate
2020-01-01
2019-03-22
2020-01-01
2018-11-30
2020-01-01

5 rows in set (0.00 sec)